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11 October 1978
SUPERSEDING
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24 May 1971

MILITARY SPECIFICATION

FORGES, FURNACES AND OVENS

(EXCLUSIVE OF SPACE HEATING AND COOKING),

PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the preservation, packing, and marking of forges, furnaces and ovens (exclusive of space heating and cooking).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.

FSC PACK

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-S-48	- Sacks, Shipping, Paper.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-D-729	- Drums: Metal, 55-Gallon (for Shipment of Noncorrosive Material).
PPP-E-911	- Excelsior, Wood, Fabricated Pads and Bulk Form.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-775	- Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles and Strainers.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition.
MIL-E-16298	- Electric Machines having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-C-52950	- Crates, Wood, Open and Covered.
MIL-B-26195	- Boxes, Wood Cleated, Skidded, Load Bearing Base.

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STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 101	- Preservation, Packaging, and Packing Materials: Test Procedures.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTOVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

3. REQUIREMENTS

3.1 First article.

3.1.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the requirements of this specification. Examination and

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tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The preproduction pack may be prepared utilizing either the preproduction model or a production forge, furnace or oven. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production forges, furnaces or ovens.

3.2 Materials. Materials shall be as specified herein and in applicable specifications. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects affecting serviceability (see 6.6).

3.3 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Disassembly. Disassembly of the forges, furnaces, and ovens shall be the minimum necessary to safeguard parts vulnerable to damage, pilferage, and loss. The extent of further disassembly shall be confined to the removal of protruding parts such as stacks, stands, and other components that would otherwise increase cubage; however, disassembly shall be limited to those parts that can be removed and reinstalled without special tools or skill. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

3.3.1.2 Matchmarking. Removed parts and mating parts on the forges, furnaces, and ovens shall be matchmarked to facilitate reassembly. Removed parts shall be identified by cloth shipping tags conforming to UU-T-81, Type A, and the marked tags shall be attached to mating parts. The tags and the printing thereon shall be waterproofed.

3.3.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116, except preservative lubricating oil specified herein shall conform to MIL-L-21260, type and grade as specified herein.

3.3.1.4 Exterior surfaces. Unpainted exterior metal surfaces of bolts, nuts, brackets, fittings, stacks, stands, and any other metal surface requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. Machined exterior

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(a) Normal voice mode. The audio frequency response at

surfaces shall be coated with Type P-6 or P-11 preservative and the coated surfaces shall be wrapped with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The wraps shall be secured with tape conforming to PPP-T-60, Type IV.

3.3.1.5 Openings. Openings into forges, furnaces, and ovens, components and assemblies that will permit the entrance of water shall be sealed with tape conforming to PPP-T-60, Type IV; large openings shall be covered with paper conforming to PPP-B-1055, Class E-1 or E-2, secured in place with tape specified herein.

3.3.1.6 Electrical components. Electrical equipment such as motors, fans, blowers, generators and other rotating electrical equipment shall be preserved in accordance with MIL-E-16298, Level A.

3.3.1.6.1 Control panels, switch panels, instrument panels, and recording instruments. All openings into control panels, switch panels, instrument panels, and recording instruments installed on equipment shall be sealed with tape conforming to PPP-T-60, Type IV. Terminals shall be coated with varnish conforming to MIL-V-13811. Control panels, switch panels, instrument panels, and recording instruments not installed on equipment shall be individually preserved in accordance with MIL-P-116, Method IIa or IIb, as applicable.

3.3.1.6.2 Capacitors. The capacitors shall be individually preserved in accordance with MIL-P-116, Method IC-2.

3.3.1.6.3 Electrical wiring. The exposed ends of wiring, openings into receptacles, junction boxes, switches, and similar items installed on the equipment shall be sealed with tape conforming to PPP-T-60, Type IV. Lengths of electric cable and wire shall be coiled to a minimum safe diameter and the coils shall be secured with tape specified herein.

3.3.1.6.4 Electric heating elements. Elements removed from the equipment shall be individually preserved in accordance with MIL-P-116, Method III, in a close-fitting box conforming to PPP-B-636, W6c, style optional. The box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.1.6.5 Pyrometers and thermocouples. Pyrometers and thermocouples shall be individually preserved in accordance with MIL-P-116, Method IC-2.

3.3.1.7 Drive belts and pulleys. Drive belts shall be removed or released from tension. Pulley faces or grooves shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, Method IC-1.

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3.3.1.8 Gears.

3.3.1.8.1 Exposed gears. Exposed gears shall be coated with Type P-1 preservative or with primer conforming to TT-P-664.

3.3.1.8.2 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housings filled to the operating level with preservative lubricating oil, Type I, grade as applicable, and the gear controls tagged to indicate: "The preservative oil contained in the gear housing is adequate for operation. Do not drain until the first required lubricant change." Gears operating on gear lubricant (SAE 80 or 90) shall have the housings filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable, and the gear controls tagged to indicate: "The lubricant contained in the gear housing is adequate for operation. Do not drain until the first required lubricant change." Gears not operating on lubricating oil (SAE 10, 30, or 50) or gear lubricant (SAE 80 or 90) shall have the housings filled to the operating level with the approved lubricant or operating oil, and the gear controls tagged to indicate: "The gear housing is filled with the approved lubricant. Do not drain." Tags specified herein shall conform to UU-T-81, Type A. The tags and printing thereon shall be waterproofed.

3.3.1.9 Drive chains.

3.3.1.9.1 Exposed chains. Exposed drive chains shall be coated with enough Type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain shall be coated with Type P-1 preservative.

3.3.1.9.2 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 3.3.1.8.2 for enclosed gears and gear housings. The housings shall be tagged as specified in 3.3.1.8.2.

3.3.1.10 Hose and hose fittings. Hose and hose fittings not installed on equipment shall be preserved in accordance with MIL-P-775, Level A. Openings into ends of hose and fittings installed on equipment shall be sealed with tape conforming to PPP-T-60, Type IV.

3.3.1.11 Oil and gas burners. The interior surfaces of oil and gas burners, piping, and valves shall be fogged with preservative lubricating oil, Type I, Grade 10. Openings into the burners, piping, nozzles, and

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jets shall be sealed with tape conforming to PPP-T-60, Type IV. Unpainted exterior surfaces of the burners, fuel ducts, piping, fittings, nozzles, and jets requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

3.3.1.12 Fuel-oil pumps. The interior surfaces of fuel-oil pumps shall be coated with preservative lubricating oil, Type I, Grade 10. Openings into the pumps shall be sealed with tape conforming to PPP-T-60, Type IV. Unpainted exterior surfaces requiring a contact preservative in accordance with MIL-P-116, shall be coated with Type P-1 preservative.

3.3.1.13 Fuel tanks. The interior surfaces of fuel tanks, if uncoated or not fabricated of corrosion-resistant material, shall be fogged with preservative lubricating oil, Type I, Grade 10. Excess preservative shall be drained and the threads of drain plugs shall be coated with the preservative oil before reinstalling. Unpainted exterior surfaces requiring the application of a preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

3.3.1.14 Crucibles and trucks. Unpainted metal surfaces of crucibles and trucks requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

3.3.1.15 Furnace coil lifts. Unpainted metal surfaces of coil lifts, guides, guide frame A-brackets, and beams requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

3.3.1.16 Drawers, shelves, and doors. Unpainted metal surfaces of drawers, shelves, and doors requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. Hinges, handles, and other fasteners requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-7 preservative. Drawers, shelves, and doors shall be secured to prevent opening and damage during shipment. Door counterweights shall be secured to prevent movement in any direction, or removed and packed separately.

3.3.1.17 Refractories.

3.3.1.17.1 Installed linings. Linings that remain installed in the basic units shall be protected with blocking and cushioning as necessary to prevent the linings from loosening or damage.

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3.3.1.17.2 Linings not installed. Refractory linings including brick, block, and shapes required for installation in the field for each basic unit shall be preserved in closing-fitting boxes conforming to PPP-B-621, Class 2, Style 2, 2-1/2, or 3, or in boxes conforming to PPP-B-601, Overseas Type, Style A or B. To prevent damage to the linings, each piece of lining shall be separated from the others and from the inside faces of the boxes by cushioning with excelsior conforming to PPP-E-911, Type I, Class B, Grade 6, or with cushioning material conforming to PPP-C-843, Type I, Class B not less than 3/8-inch thick. Strapping shall not be required. The gross weight of each box shall not exceed 200 pounds. When specified, refractory linings required for each basic unit in such quantities as to prohibit packing with the basic unit shall be preserved in nailed wood boxes specified in this paragraph. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A. The gross weight of each box shall not exceed 500 pounds. The boxes shall be shipped without overpacking at the same time as the basic unit for which intended.

3.3.1.17.3 Dry refractory and portland cement. Dry refractory (fire clay) shall be preserved in quantities of 60 pounds plus 1 pound minus 0 in Type IV sacks conforming to UU-S-48, Level A. Portland cement shall be preserved in quantities of 94 pounds plus 1 pound minus 0 (0.48 cubic foot by absolute volume), or a lesser quantity when required for an installation, in Type IV sacks conforming to UU-S-48, Level A. The bags shall be closed in accordance with the bag specification.

3.3.1.17.4 Liquid paste and plastic cement. Wet-type pastes and cements shall be preserved in the quantities required for each basic unit in 1-gallon or 5-gallon cans, as applicable, conforming to PPP-C-96, Type V, Class 1, Plan B, or when applicable, in 55-gallon metal drums conforming to PPP-D-729, Type IV.

3.3.1.17.5 Sand. Sand and similar materials shall be preserved in quantities of 80 pounds plus 1 pound minus 0 in Type II sacks conforming to UU-S-48, Level A.

3.3.1.18 Insulation.

3.3.1.18.1 Blanket. Blanket insulation in rolls shall be individually preserved in accordance with MIL-P-116, Method III. The containers shall be close-fitting boxes conforming to PPP-B-636, V3c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

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3.3.1.18.2 Board or block. Board or block insulation of like description shall be preserved in accordance with MIL-P-116, Method III. The containers shall be close-fitting boxes conforming to PPP-B-636, W6c, style optional. Quantities shall not exceed 70 pounds gross weight. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.1.18.3 Loose fill. Loose fill insulation shall be preserved in Type II sacks conforming to UU-S-48, Level A.

3.3.1.19 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, Level A.

3.3.1.20 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1.

3.3.1.21 Repair parts. Unless otherwise specified repair parts shall be preserved in accordance with MIL-P-116.

3.3.1.22 Fire extinguishers. Unpainted exterior surfaces of the fire extinguishers requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The extinguishers shall be placed and secured in the mounting brackets. When mounting brackets are not provided the fire extinguishers shall be individually preserved in a close-fitting box conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

3.3.1.23 Consolidation.

3.3.1.23.1 Components. Components and assemblies for each forge, furnace, or oven, not installed on the equipment, shall be consolidated in close-fitting boxes conforming to PPP-B-636, Class Weather Resistant, style optional; or in boxes conforming to PPP-B-640, Class 2, style optional. Strapping shall not be required. The gross weight of each box conforming to PPP-B-636 shall not exceed the limitation of the box specification. The gross weight of each box conforming to PPP-B-640 shall not exceed 200 pounds.

3.3.1.23.2 Repair parts, tools and technical publications. The preserved repair parts, tools and technical publications required for each forge, furnace or oven shall be consolidated in quantities not to exceed the weight limitation of boxes conforming to PPP-B-636, Class Weather Resistant, style optional.

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3.3.2 Level B. Unless otherwise specified (see 6.4), Level B preservation shall be as follows.

3.3.2.1 Disassembly and matchmarking. Disassembly and matchmarking shall be as specified in 3.3.1.1 and 3.3.1.2 for Level A, except tags used for matchmarking need not be waterproofed.

3.3.2.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116, except preservative lubricating oil specified herein shall conform to MIL-L-21260, type and grade as specified.

3.3.2.3 Exterior surfaces. Unpainted exterior metal surfaces of bolts, nuts, fittings, stacks, stands, or any other metal surface requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. Machined surfaces shall be coated with Type P-6 preservative and wrapped with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The wraps shall be secured with tape conforming to PPP-T-60, Type IV.

3.3.2.4 Electrical components. Electrical equipment such as motors, fans, blowers, generators, and other rotating electrical equipment shall be preserved in accordance with MIL-E-16298, Level A, using the alternate method as specified for assembled machines.

3.3.2.4.1 Control panels, switch panels, instrument panels, and recording instruments. Control panels, switch panels, instrument panels, and recording instruments not installed on equipment shall be individually preserved in accordance with MIL-P-116, Method IC, submethod as applicable.

3.3.2.4.2 Capacitors. The capacitors shall be individually preserved in accordance with MIL-P-116, Method IC, submethod as applicable.

3.3.2.4.3 Electrical wiring. Lengths of electric cable and wire shall be coiled to not less than 10 times their diameter and the coils shall be secured with tape conforming to PPP-T-60, Type IV.

3.3.2.4.4 Electric heating elements. Electric heating elements removed from the equipment shall be individually preserved in accordance with MIL-P-116, Method III.

3.3.2.4.5 Pyrometers and thermocouples. Pyrometers and thermocouples shall be individually preserved in accordance with MIL-P-116, Method III.

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3.3.2.5 Drive belts. Drive belts shall be released from tension.

3.3.2.6 Gears.

3.3.2.6.1 Exposed gears. Exposed gears shall be coated with Type P-1 preservative.

3.3.2.6.2 Enclosed gears. The gear housings shall be filled to the operating level with the lubricant required for operation. A tag shall be attached indicating: "THE HOUSING IS FILLED TO THE OPERATING LEVEL WITH THE LUBRICANT REQUIRED FOR OPERATION. DO NOT DRAIN."

3.3.2.7 Drive chains.

3.3.2.7.1 Exposed chains. Exposed drive chains shall be coated with enough Type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the pins, rollers, and bushings.

3.3.2.7.2 Enclosed chains. Enclosed chain housings shall be filled to the operating level with the lubricant required for operation. A tag shall be attached indicating: "THE HOUSING IS FILLED TO THE OPERATING LEVEL WITH THE LUBRICANT REQUIRED FOR OPERATION. DO NOT DRAIN."

3.3.2.8 Hose and hose fittings. Hose and hose fittings not installed on equipment shall be preserved in accordance with MIL-P-775, Level A.

3.3.2.9 Oil and gas burners, fuel-oil pumps, and fuel tanks. Interior surfaces of oil and gas burners, fuel-oil pumps, and uncoated and noncorrosion-resistant fuel tanks shall be coated with preservative lubricating oil Type I, Grade 10. Excess preservative shall be drained. Fuel tank drain plugs shall be coated with the preservative oil and reinstalled.

3.3.2.10 Crucibles, trucks, and coil lifts. Unpainted metal surfaces of crucibles, trucks, and coil lifts shall be coated with Type P-1 preservative.

3.3.2.11 Drawers, shelves, and doors. Drawers, shelves, and doors shall be secured to prevent opening and damage during shipment. Door counterweights shall be secured to prevent movement in any direction, or removed and packed separately.

3.3.2.12 Refractories.

3.3.2.12.1 Installed linings. Linings that remain installed in the basic units shall be protected with blocking and cushioning as necessary to prevent the linings from loosening or damage.

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3.3.2.12.2 Linings not installed. Refractory linings, including brick, block, and shapes required for installation in the field for each basic unit, shall be preserved in close-fitting boxes conforming to PPP-B-621, Class 1, style optional, or in boxes conforming to PPP-B-601, Domestic Type, style optional. To prevent damage to the linings, each piece of lining shall be separated from the others and from the inside faces of the boxes by cushioning with excelsior conforming to PPP-E-911, Type I, Class B, Grade 6, or with cushioning material conforming to PPP-C-843, Type I, Class B, not less than 3/8-inch thick. Strapping shall not be required. The gross weight of each box shall not exceed 200 pounds. When specified, refractory linings required for each basic unit in such quantities as to prohibit packing with the basic unit shall be preserved in nailed wood boxes specified in this paragraph. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable, Finish A. The gross weight of each box shall not exceed 500 pounds. The boxes shall be shipped without overpacking at the same time as the basic unit for which intended.

3.3.2.12.3 Dry refractory and portland cement. Dry refractory (fire clay) shall be preserved in quantities of 60 pounds plus 1 pound minus 0 in Type IV sacks conforming to UU-S-48, Level B. Portland cement shall be preserved in quantities of 94 pounds plus 1 pound minus 0 (0.48 cubic foot by absolute volume), or a lesser quantity when required for an installation, in Type IV sacks conforming to UU-S-48, Level B. The bags shall be closed in accordance with the bag specification.

3.3.2.12.4 Liquid paste and plastic cement. Wet-type pastes and cements shall be preserved in the quantities required for the basic unit in 1-gallon or 5-gallon cans, as applicable, conforming to PPP-C-96, Type V, Class 1, Plan B, or when applicable, in 55-gallon metal drums conforming to PPP-D-729, Type III.

3.3.2.12.5 Sand. Sand and similar materials shall be preserved in quantities of 80 pounds plus 1 pound minus 0 in Type II sacks conforming to UU-S-48, Level B.

3.3.2.13 Insulation.

3.3.2.13.1 Blanket. Blanket insulation in rolls shall be individually preserved in accordance with MIL-P-116, Method III. The containers shall be close-fitting boxes conforming to PPP-B-636, Domestic Class, style optional.

3.3.2.13.2 Board or block. Board or block insulation of like description shall be preserved in accordance with MIL-P-116, Method III. The containers shall be boxes conforming to PPP-B-636, Domestic Class, style optional.

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3.3.2.13.3 Loose fill. Loose fill insulation shall be preserved in Type II sacks conforming to UU-S-48, Level B.

3.3.2.14 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, Level A.

3.3.2.15 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1.

3.3.2.16 Repair parts. Unless otherwise specified repair parts shall be preserved in accordance with MIL-P-116.

3.3.2.17 Fire extinguishers. Fire extinguishers shall be placed and secured in the mounting brackets.

3.3.2.18 Consolidation.

3.3.2.18.1 Components. Components and assemblies for each forge, furnace, or oven, if not installed on the equipment, shall be consolidated in close-fitting boxes conforming to PPP-B-636, weather-resistant, style optional; or in boxes conforming to PPP-B-640, Class 2, style optional. Strapping shall not be required. The gross weight of each box conforming to PPP-B-636 shall not exceed the weight limitation of the box specification. The gross weight of each box conforming to PPP-B-640 shall not exceed 200 pounds.

3.3.2.18.2 Repair parts, tools and technical publications. The preserved repair parts, tools and technical publications required for each forge, furnace or oven shall be consolidated in quantities not to exceed the weight limitation of boxes conforming to PPP-B-636, Class weather resistant, style optional.

3.3.3 Commercial. The forges, furnaces, ovens, components, and the repair parts, tools, and technical publications shall be preserved in accordance with MIL-STD-1188 (see 6.5). When linings remain installed in the basic unit, the necessary blocking and bracing shall be provided to prevent the linings from becoming loose or damaged during shipment.

3.4 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 Forges, furnaces, and ovens, weight not to exceed 1,000 pounds. Each complete forge, furnace, or oven, with components, and repair parts comprising a complete unit, preserved as specified in 3.3, shall be packed

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in a close-fitting box conforming to PPP-B-621, Class 2, Style 2, 2-1/2, or 3; or in a box conforming to PPP-B-601, Overseas Type, Style A or B or skidded boxes conforming to MIL-B-26195, Type II, Style A, class optional, with plywood superstructure. Contents of boxes shall be cushioned, blocked, and braced in accordance with MIL-STD-1186. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

3.4.1.2 Forges, furnaces, and ovens, weight exceeding 1,000 pounds. Forges, furnaces, or ovens, with components, and repair parts comprising a complete unit of equipment, preserved as specified in 3.3, with weight exceeding 1,000 pounds, shall be packed in a crate conforming to MIL-C-104, Type I or II, Class 1 or 2, Style a or for loads not exceeding 2500 pounds in a box conforming to MIL-B-26195, Type II, Style A, class optional, with plywood superstructure. Blocking, bracing, and anchoring shall be in accordance with the appendix to the crate specification and in accorddance with MIL-STD-1186.

3.4.2 Level B. The complete units of equipment, preserved as specified in 3.3, shall be packed as specified for Level A, except that wood boxes shall conform to MIL-B-26195, Type I, Style A, class optional, plywood superstructure, PPP-B-621, Class 1, style as applicable; wood cleated-plywood boxes shall conform to PPP-B-601, Domestic Type, style optional; and crates shall conform to MIL-C-3774, Type I, Style A, or MIL-C-52950, Type V, Style A. Contents of the crates shall be waterproofed with the full shroud in accordance with the appendix to the applicable crate specification. Strapping shall be Finish A.

3.4.3 Commercial. The forges, furnaces, ovens, components, and the repair parts, tools and technical publications shall be packed in accordance with MIL-STD-1188.

3.5 Marking.

3.5.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

3.5.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in

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the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined as specified in 4.4.4. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Unless otherwise specified (see 6.2), the preproduction pack for Levels A and B shall be testedin accordance with 4.3.2.1 or 4.3.2.2, as applicable to the type of container (box or crate). Failure of any test shall be cause for rejection. Any shifting of or visible damage to the contents or loosening, breaking, or displacement of anchoring, blocking, or bracing within the container shall constitute failure of the test.

4.3.2.1 Boxed forges, furnaces, or ovens. Boxed forges, furnaces, or ovens not exceeding 200 pounds shall be subjected to the free-fall drop test in accordance with FED. TEST METHOD STD No. 101, Test Method 5007. Boxed forges, furnaces, or ovens exceeding 200 pounds shall be subjected to the pendulum-impact test in accordance with FED. TEST METHOD STD. No. 101, Test Method 5012.

4.3.2.2 Crated forges, furnaces, or ovens. Crated forges, furnaces, or ovens shall be subjected to the guided-impact test (railcar) in Appendix A to MIL-STD-1186. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of between 10 miles and 11 miles per hour.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. Inspection shall be in two stages as follows:

(a) The first stage shall include inspection of materials, methods, containers and markings inside the shipping container.

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(b) The second stage shall include inspection of materials and marking after closing the shipping container.

4.4.2 Unit of product. For the purpose of inspection, a completed pack of any one level prepared for shipment shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in Table I. AQL shall be 2.5 percent defective.

Table I. Examination

	Defect	Level A	Level B
101.	Materials, containers, or methods not as specified. Each incorrect material, container, or method shall constitute one defect (3.2, 3.3, 3.4).	X	X
102.	Disassembly not as specified (3.3.1.1, 3.3.2.1).	X	X
103.	Matchmarking not as specified (3.3.1.2, 3.3.2.1).	X	X
104.	Tags used for matchmarking not waterproofed (3.3.1.2).	X	-
105.	Bolts, nuts, pins, screws, and washers removed during disassembly not reinstalled in mating parts (3.3.1.1, 3.3.2.1).	X	X
106.	Machined surfaces not coated with preservative and wrapped as specified (3.3.1.4, 3.3.2.3).	X	X
107.	Openings that will permit the entrance of water not sealed as specified (3.3.1.5).	X	-
108.	Openings into control panels, switch panels, instrument panels, and recording instruments installed on equipment not sealed as specified (3.3.1.6.1).	X	-
109.	Exposed ends of wiring, openings into receptacles, junction boxes, and switches not sealed as specified (3.3.1.6.3).	X	-
110.	Lengths of electrical cable and wire not coiled and the coils secured (3.3.1.6.3, 3.3.2.4.3).	X	X
111.	Drive belts not removed or released from tension as applicable (3.3.1.7, 3.3.2.5).	X	X
112.	Gear housings not tagged and marked as specified (3.3.1.8.2, 3.3.2.6.2).	X	X
113.	Chain housings not tagged and marked as specified (3.3.1.9.2, 3.3.2.7.2).	X	X

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Table I. Examination (Cont'd)

Defect		Level A	Level B
114.	Openings into ends of hose and openings into fittings not sealed as specified (3.3.1.10).	X	-
115.	Interior surfaces of oil and gas burners not coated with preservative as specified (3.3.1.11, 3.3.2.9).	X	X
116.	Interior surfaces of fuel-oil pumps not coated with preservative as specified, and openings not sealed as specified (3.3.1.12, 3.3.2.9).	X	X
117.	Interior surfaces of fuel tanks and threads of drain plugs not coated with preservative as specified (3.3.1.13, 3.3.2.9).	X	X
118.	Excess preservative not drained from fuel tanks and the drain plugs not reinstalled (3.3.1.13, 3.3.2.9).	X	X
119.	Surfaces of drawers, shelves, and doors not coated with preservative as specified (3.3.1.16).	X	-
120.	Drawers, shelves, and doors not secured to prevent opening during shipment (3.3.1.16, 3.3.2.11).	X	X
121.	Installed linings not blocked and cushioned to prevent loosening (3.3.1.17.1, 3.3.2.12.1).	X	X
122.	Preserved linings not separated with excelsior or cushioning as specified (3.3.1.17.2, 3.3.2.12.2).	X	X
123.	Strapping not as specified (3.3.1.17.2, 3.3.2.12.2, 3.4.1.1).	X	X
124.	Bag closure for dry refractory and portland cement not as specified (3.3.1.17.3, 3.3.2.12.3).	X	X
125.	Fire extinguishers not secured in the mounting brackets (3.3.1.22, 3.3.2.17).	X	X
126.	Contents of open crates not waterproofed as specified (3.4.2).	-	X
127.	Marking illegible, incomplete, incorrect or missing (3.5).	X	X

5. PACKAGING

(Not applicable)

6. NOTES

6.1 Intended use. It is intended that this specification be used for reference in Section 5 of commodity and equipment specifications or for direct reference in contracts, or for reference in the preparation of packaging data sheets.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a preproduction pack is not required.
(c) Time frame required for submission of preproduction pack (see 3.1.1).
(d) Degree of preservation and degree of packing required (see 3.3 and 3.4). Level B preservation is intended to provide economical but limited protection and should be specified only when it is determined the forge, furnace, or oven will be held in covered storage.
(e) Requirements if other than as specified (3.3.1.21 and 3.3.2.16).
(f) When linings not installed are to be packed for separate shipment (see 3.3.1.17.2 and 3.3.2.12.2).
(g) When testing of the preproduction pack is not required (see 4.3.2).
(h) When forges, furnaces, or ovens shall be subjected to operational tests (see 4.3.2.3).
(i) When Finish A strapping is required (see 3.3.1.17.2 and 3.4.1.1).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the forges, furnaces, and ovens in accordance with this specification.

6.4 Level B preservation. Level B preservation (see 3.3.2), has been developed for use by the Army and Air Force for the specific conditions specified in 6.2. For Navy procurements, details of Level B preservation, when required, should be developed at the time of procurement when all conditions are known.

6.5 Level A for repair parts, tools, and publications. When forges, furnaces and ovens are not preserved Level A, but it is desired that repair parts, tools and technical publications be preserved Level A, the contract should state: "Except that repair parts, maintenance tools and technical publications shall be preserved, and consolidated in accordance with the Level A requirements (see 3.3.3)."

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6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to the material specifications or preparing activity for recommended disposability methods.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Army - SM
Navy - SH

User activity:
Navy - MC

Preparing activity:
Army - ME

Project PACK-0573

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-F-3296D Forges, Furnaces and Ovens (Exclusive of Space Heating and Cooking), Packaging Of

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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